

Sentence Structure of Technical Writing

“The fundamental purpose of scientific discourse is not the mere presentation of information and thought but rather *its actual communication*. It does not matter how an author might be to have converted all the right data into sentences and paragraphs; *it matters only whether a large majority of the reading audience accurately perceives what the author had in mind.*”

--George Gopen and Judith Swan
The Science of Scientific Writing

Good Tech Writers Practice

- Planning**
- Clarity**
- Brevity**
- Simplicity**
- Word Choice**
- Active Voice**
- Committing to Writing as a Process**

Planning: Before You Begin

Identify your audience and their expectations

Know your purpose

Know your material

Understand the writing task at hand

Organize your thoughts and materials

Budget adequate time to write, review, revise and edit

Clarity: Avoid Jargon

Jargon: a vocabulary particular to a place of work (*abbreviations, slang*)

• ***Audience familiarity with the topic determines appropriate use of jargon***

Ex. 1: For the first year, the links with SDPC and the HAC were not connected, and all required OCS input data were artificially loaded. Thus CATCH22 and MERWIN were not available.

Ex. 2: Because some of the links in the computer system were not connected the first year, we could not run all the software codes.

Clarity: Define the Unfamiliar

If you must abbreviate, define the term in its first occurrence, and ***put abbreviations in parentheses***

Ex: Edgartown Great Pond (EGP) is a vital body of water. Unfortunately, due to an unpredictable influx of saltwater, the delicate ecosystem is in danger of destabilizing.

• ***Italicize first occurrence of unfamiliar terms and define them right away***

Ex: *Retina is a light-sensitive tissue, found at the back of the eye, that converts light impulses to nerve impulses.*

Brevity: Use Words Efficiently

Never use two words when one word will do.

Ex. 1: The relationship between the nature of salt water to fresh water in the Edgartown Great Pond that fluctuates often is extremely important to everyone including scientists, residents, and environmentalists on Martha's Vineyard.

Ex. 2: The fluctuating salinity of EGP concerns many environmentalists, scientists, and residents

Brevity: Less Is More

Pare your language down to the essential message you want to get across to your readers:

Ex: Earthquakes can occur at predictable intervals along a given fault segment. Depending on the length and slip in each mainshock, the exact interval can vary by a factor of two. The southern segment of the San Andreas fault has an interval of 145 years, plus or minus a few decades

Brevity: Most Important First

Place key *information in the main clause*

Ex. 1: Despite winning the game, the Patriots made several errors in the first half.

Ex. 2: Despite making several errors in the first half, the Patriots won the game.

Ex. 3: The Patriots won the game, despite making several errors in the first half

Brevity: Remove Redundancy

Combine overlapping sentences when possible

Ex. 1: Water quality in Hawk River declined in March. This decline occurred because of the heavy rainfall that month. All the extra water overloaded Tomlin county's water treatment plant.

Ex. 2: Water quality in Hawk River declined in March because heavy rainfalls overloaded Tomlin County water treatment plant.

Simplicity: Use Details Wisely

Specific details are desirable, but be careful to ***balance detail with audience needs for clarity—significance is more important.***

Ex. 1: The average house in the area has a radon level of 0.4 picocuries per liter.

Ex. 2: The average house in the area has a radon level of 0.4 picocuries per liter, which is considered low by the EPA [Lafavore, 1987]. Levels between 20 and 200 picocuries per liter are considered high, and levels above 200 picocuries per liter are considered dangerous. For reference, the average radon level in out

Simplicity: Use Details Wisely

Many engineers want to provide as much specific detail as possible, but this can come at the expense of readers understanding and their main point

Ex. 1: The number of particular hydrocarbon combinations in our study is enormous. For example, the number of possible $C_{20}H_{42}$ is 366,319 and the number of $C_{40}H_{82}$ is 62,491,178,805,831.

Ex. 2: The number of hydrocarbon combinations in our study is enormous. For example, the number of possible $C_{40}H_{82}$ is over 60 trillion.

Language: Needless Complexity

Category

Example

Substitute

noun	Utilization functionaliot y	Use feature
verb	Facilitate finalize	Cause end
adjective	Aforementio ned invidualized	Mentioned individual

Language: Abstraction

Avoid too many abstract nouns

Ex. 1: The existing nature of Mount St. Helens' volcanic ash spew age was handled through the applied use of computer modeling capabilities.

Ex. 2: With Cray computers, we modeled how much ash spewed from Mount St. Helens.

Language: Needless Words

(already) existing
at(the) present (time)
(basic) fundamentals
(completely) eliminate
(continue to) remain
currently(being)
(currently) underway
(empty) space
had done(previously)
introduced(a new)
mix(together

never (before)
none(at all)
now(at this time)
period(of time)
(private) industry
(separate) entities
start(out)
write(out)
(still) persists

Language: Ambiguity

Choose words whose meanings are ***clear***

Ex. 1: T cells, rather than B cells, appeared as the lymphocytes migrated to the thymus gland.

Ex. 2: T cells, rather than B cells, appeared because the lymphocytes migrated to the thymus gland

Language: Ambiguity

Order the words in your sentences carefully

Ex. 1: In low water temperatures and high toxicity levels of oil, we tested how well the microorganisms survived.

Ex. 2: We tested how well the microorganisms survived in low water temperatures and high toxicity levels of oil.

Language: Ambiguity

Do not overuse pronouns—particularly “it ”and “this”—because it is often difficult to identify the antecedent

Ex: Because the receiver presented the radiometer with a high-flux environment, it was mounted in a silver-plated stainless steel container. Because the receiver presented the radiometer with a high-flux environment, it was mounted in a silver-plated stainless steel container.

Language: Weak vs. Strong

Avoid too many “to be” verbs

“is” “was” “were” “has been” “have been”

• ***Avoid excess words, which slow comprehension of the main point***

made arrangements for	arranged
made the decision	decided
made the measurement of	measured

performed the development of	developed
is working as expected	works as expected

Active Voice: Strong Verbs

Technical writers want to communicate as efficiently as possible, and ***active voice is more straightforward and is stronger than passive voice***

Ex 1: The feed through was composed of a sapphire optical fiber, which was pressed against the pyrotechnic that was used to confine the charge.

Ex 2: The feed through contained a sapphire optical fiber, which pressed against the pyrotechnic that contained the charge.

Active Voice: Natural Sound

When in doubt, ***read passages out loud to determine the natural sound****

Ex 1: A new process for eliminating nitrogen oxides from diesel exhaust engines is presented. Flow tube experiments to test this process are discussed. A chemical reaction scheme to account for this process is proposed.

Ex 2: We present a new process for eliminating nitrogen oxides from the exhaust of diesel engines. To test this process, we performed experiments in flow tubes. To explain this process, we developed a scheme of chemical reactions.

*always defer to your professor, your journal, or your company style guide for use of “I” and “we” in technical paper

Writing Is a Process

Good writing doesn't happen overnight; ***it requires planning, drafting, rereading, revising, and editing. Learning and improvement requires self-review, peer-review, subject-matter expert feedback, and practice. There are no shortcuts; practice makes perfect!***

To summarize

- *Plan your project before you begin drafting.*
- *Understand basic qualities of good technical writing; use the examples presented to guide you in your writing and revising process.*
- *Good writing is a habit that takes time to develop; practice makes perfect.*